

CONCEPT PAPER: ONLINE RETAIL MANAGEMENT SYSTEM

1. Introduction

With the rapid growth of digital technology, businesses are increasingly moving from traditional physical stores to online platforms. An online retail system allows customers to browse products, place orders, and make payments conveniently from anywhere. This concept focuses on designing a database system using an ERD to ensure efficient data management.

2. Problem Statement

Many businesses face challenges in managing customer orders, tracking products, and handling payments efficiently. Manual systems lead to errors and poor customer experience. A structured system is needed to manage customers, orders, products, and carts.

3. Objectives

Main Objective: Design a database system for an online retail platform

Specific Objectives:

- Model customer and order management
- Organize products and categories
- Manage shopping cart
- Track payments

4. System Description

The system includes Customer, Product, Category, Cart, CartItem, Order, OrderItem, and Payment entities. These work together to support online retail operations.

5. Relationships

- Customer has a Cart
- Customer places an Order
- Order contains OrderItems
- Product belongs to Category
- Cart contains CartItems
- Order has a Payment

6. Significance

Improves data organization, reduces redundancy, enhances accuracy, and improves customer experience.

7. Conclusion

The system provides a reliable and efficient structure for managing online retail operations.

8. Future Improvements

- Delivery tracking
- Mobile money integration
- Reviews and ratings.